TRAINING NEEDS ASSESSMENT

CFITES Performance Objective Writing Competency

Future Aircrew Training (FAcT) Programme — SkyAlyne Canada / CAE

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EXECUTIVE SUMMARY

This Training Needs Assessment (TNA) was conducted to determine whether a performance gap exists in CFITES-aligned Performance Objective (PO) writing competency among courseware developers and instructional designers assigned to the Future Aircrew Training (FAcT) programme. The assessment examined the current state of PO writing practice against the standard required by the CFITES framework (A-P9-050-000/PT-006) and identified a consistent, programme-wide gap driven by insufficient foundational training in CFITES PO construction methodology.

The gap is consequential: poorly specified POs propagate downstream, misaligning courseware, assessment instruments, and evaluation criteria across the FAcT lifecycle. The assessment recommends a targeted eLearning intervention — estimated at 15–20 minutes of learner time — focused exclusively on PO writing at the three-

component level (Action, Conditions, Standard). This document provides the evidentiary basis for that recommendation and the instructional parameters that should govern the resulting training design.

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1. ORGANIZATIONAL AND PROGRAMME CONTEXT

1.1 Programme Overview

The Future Aircrew Training (FAcT) programme is a Government of Canada initiative delivered under a Public-Private Partnership (P3) contract between SkyAlyne Canada (a joint venture of CAE and KF Aerospace) and the Royal Canadian Air Force (RCAF). The programme provides comprehensive training for all fixed-wing RCAF aircrew roles, integrating live flight, full-motion simulation, and ground school instruction across multiple aircraft types, including the Pilatus PC-21 and Grob G 120TP.

Training under FAcT is governed by the Canadian Forces Individual Training and Education System (CFITES), the 14-volume Quality Control framework that defines how all CAF individual training is designed, developed, conducted, and evaluated. Compliance with CFITES is a contractual obligation of the FAcT P3 agreement and a prerequisite for DND acceptance of FAcT courseware.

1.2 Instructional Design Function Within FAcT

SkyAlyne employs a team of courseware developers and instructional designers (IDs) responsible for producing and maintaining the ground school curriculum, simulator training packages, and associated job aids that support FAcT instructor-led training. These personnel work closely with Subject Matter Experts (SMEs) — primarily qualified flying instructors (QFIs) and technical training officers — to translate operational competency requirements into structured learning experiences.

The ID team operates at the intersection of the CFITES Design and Development phases. Their primary inputs are Task Lists and Performance Objectives produced during the Analysis phase; their primary outputs are Training Plans, Lesson Plans, courseware, and formative/summative assessment instruments. The quality of POs they receive from — or write in collaboration with — SMEs directly determines the quality and CFITES-compliance of all downstream deliverables.

1.3 Triggering Event for This Assessment

This TNA was initiated following a CFITES compliance audit of FAcT Phase 1 ground school courseware conducted by DND's Directorate of Training and Education in Q4 2025. The audit identified a recurring pattern of non-conformant POs across multiple courseware packages — specifically, POs with vague or unobservable action verbs, absent or underspecified conditions statements, and subjective standards that could not be applied consistently across evaluators.

The audit resulted in a Corrective Action Request (CAR) requiring SkyAlyne to demonstrate that its courseware development workforce has been trained to CFITES PO writing standards. This TNA was commissioned to establish the baseline, characterize the gap, and recommend an appropriate intervention.

2. ASSESSMENT METHODOLOGY

2.1 Assessment Framework

This TNA followed the CFITES Needs Assessment phase methodology as described in CFITES Volume 2 (Analysis of Occupational Requirements) and is aligned with the Training Needs Analysis model recommended by the Treasury Board Secretariat for federally contracted training programmes. The assessment proceeded in three stages:

Stage	Activity	Purpose	Timing
1 — Data Collection	Document review, SME interviews, direct observation	Establish current-state performance baseline	Weeks 1–2
2 — Gap Analysis	Comparison of current-state evidence against CFITES standard	Quantify and characterize the performance gap	Week 3
3 — Causal Analysis	Structured interviews, root-cause mapping	Identify enabling and inhibiting factors driving the gap	Week 3–4

2.2 Data Sources

The following data sources were consulted during the assessment:

CFITES documentary review	CFITES Volumes 2, 4, and 6; A-P9-050-000/PT-006 (Vol. 6 — Conduct of Instructional Programmes); FAcT Programme Training Plan (PTP) v2.1; DND Qualification Standard templates
Courseware audit findings	Q4 2025 DND compliance audit report; 23 FAcT POs sampled across 6 ground school modules; evaluator scoring rubric from the audit
SME interviews	Semi-structured interviews with 4 QFIs (SMEs), 3 courseware developers, 1 Training Manager, and 1 DND Standards Officer (n=9 total)
Direct observation	Two courseware development team meetings observed; one PO writing workshop facilitated by a QFI observed
Artefact analysis	Examination of 15 POs written by current team members without SME input; comparison against 8 POs from an earlier, externally-contracted courseware package that passed DND review

2.3 Assessment Limitations

The following limitations are noted:

- The artefact sample (n=23 POs) is drawn from Phase 1 ground school only; PO writing practice in simulator training packages was not assessed in this cycle.
- Interview data is self-reported; social desirability bias may have led some participants to understate their difficulty with CFITES requirements.
- One QFI SME declined to participate in the interview phase, citing workload constraints.
- No pre/post training measurement was possible at this stage; the gap characterization is based on artefact analysis and structured observation only.

3. CURRENT STATE ANALYSIS

3.1 Target Population

The primary target population for this assessment is the SkyAlyne FAcT courseware development team. Secondary populations include QFI SMEs who contribute to or review PO drafts. The table below profiles the primary population:

Characteristic	Description
Population size	7 courseware developers / instructional designers (full-time, on-site, Portage la Prairie MB and Ottawa ON)
Educational background	Mix of adult education certificates, B.Ed. degrees, and one MA (Educational Technology). No team member holds a formal CFITES qualification.
Prior CFITES training	3 of 7 report having attended a one-day CFITES orientation briefing during FAcT programme onboarding (2023). None have completed a formal CFITES PO writing course.
Aviation domain knowledge	Variable. Two team members have prior military training experience (Army, not RCAF). Five have no prior defence sector experience.
Authoring tool proficiency	All 7 proficient in Microsoft Office. 5 of 7 have used at least one eLearning authoring tool (Articulate 360 n=2, Adobe Captivate n=3, Lectora n=1). 2 are new to eLearning authoring.
Primary language	English (all). French (second language, limited working proficiency): 3 of 7.

3.2 Required Performance Standard

The CFITES standard for a compliant Performance Objective is defined in CFITES Volume 4 (Design of Instructional Programmes) and requires that every PO contain the following three components, each meeting specified criteria:

Component	CFITES Requirement	Compliance Criteria
Action (Performance Statement)	An observable, measurable action verb + object describing what the learner must perform	Verb must be observable (eliminates: understand, know, appreciate, be aware of) Verb must map to a CFITES cognitive level (Recall / Apply / Problem-Solve) Verb must describe learner behaviour, not instructor or training activity
Conditions (Conditions Statement)	A specification of the resources, restrictions, and environment under which the performance will occur	Must identify resources provided (equipment, references, data) Must identify any denied resources or restrictions where relevant Must specify the training setting or environment Must be specific enough to replicate evaluation conditions independently

Component	CFITES Requirement	Compliance Criteria
Standard (Standard Statement)	An objective, measurable criterion defining minimum acceptable performance	Must use quantifiable terms: accuracy, time, compliance with a reference document, or score threshold Must not use subjective language: satisfactory, appropriate, good, proper Must be specific enough for two evaluators to reach the same pass/fail decision independently

3.3 Current Performance Evidence

Analysis of the 23 POs sampled from the DND audit, supplemented by direct artefact analysis of 15 team-authored POs, produced the following findings:

3.3.1 Action Component — Error Rate and Error Types

Finding	Frequency	Example from Sample
Non-observable verb used as primary action	61% of sampled POs (14/23)	'The student will understand the principles of instrument scanning.'
Vague verb that resists level mapping	35% of sampled POs (8/23)	'The student will know the emergency drill sequence.'
Action describes instructor activity, not learner performance	13% of sampled POs (3/23)	'Students will be introduced to pre-flight inspection procedures.'
Compliant action verb	26% of sampled POs (6/23)	'The student pilot will execute the engine failure drill…'

3.3.2 Conditions Component — Error Rate and Error Types

Finding	Frequency	Example from Sample
Conditions entirely absent	48% of sampled POs (11/23)	PO proceeds directly from learner subject to action verb with no conditions clause.
Conditions present but vague	30% of sampled POs (7/23)	'Given appropriate training materials, the student will…'
Training environment unspecified	52% of sampled POs (12/23)	No indication of whether task is performed in classroom, simulator, or aircraft.
Compliant conditions statement	22% of sampled POs (5/23)	'Given a PC-21 full-motion simulator and applicable QRH…'

3.3.3 Standard Component — Error Rate and Error Types

Finding	Frequency	Example from Sample
Standard entirely absent	39% of sampled POs (9/23)	PO ends after the action clause with no standard stated.
Subjective standard (cannot be evaluated consistently)	44% of sampled POs (10/23)	'…to a satisfactory standard.' / '…demonstrating competent airmanship.'
Standard present and measurable	17% of sampled POs (4/23)	'…within 15 seconds of warning light activation, with no critical steps omitted.'

Only 4 of 23 sampled POs (17%) contained all three CFITES-compliant components. The remaining 83% contained at least one deficiency serious enough to require revision before the PO could be used to drive courseware development or formative assessment design.

4. GAP ANALYSIS

4.1 Gap Summary

The performance gap is characterized as follows:

Dimension	Current State	Required State	Gap Severity
Action verb selection	61% non-observable verbs; no consistent application of CFITES cognitive level taxonomy	100% observable, measurable verbs mapped to appropriate CFITES level	HIGH
Conditions specification	48% absent; 30% vague; 22% compliant	100% specific, replicable conditions statements	HIGH
Standard specification	39% absent; 44% subjective; 17% compliant	100% objective, measurable standards	HIGH
Full PO compliance (all 3 components)	17% of POs fully compliant	100% of POs fully compliant for DND acceptance	CRITICAL
CFITES framework literacy (conceptual)	Variable; most team members cannot articulate all three PO components unprompted	All team members able to identify, diagnose, and correct all three components	HIGH

4.2 Downstream Impact of the Gap

The PO writing gap is not an isolated quality issue. Within the CFITES lifecycle, a non-compliant PO produces a cascade of downstream misalignment:

CFITES Phase	Impact of Non-Compliant PO
Design	Training Plan cannot be accurately scoped; content objectives are misaligned to the PO; media/method selection is arbitrary rather than evidence-driven.
Development	Courseware content is developed to an unclear target; scenario-based learning cannot be calibrated to a measurable performance standard; formative quiz items cannot be reliably mapped to the PO.
Conduct	Instructors cannot use the PO to communicate clear lesson expectations to learners; "what good looks like" is not defined for in-lesson coaching.
Evaluation	Summative evaluation instruments cannot be built against a non-measurable standard; inter-rater reliability of evaluator scoring is compromised; DND acceptance criteria cannot be met.

5. CAUSAL ANALYSIS

5.1 Root Causes of the Performance Gap

Structured interviews and observation identified four primary causes of the PO writing gap. These are not motivational or attitudinal in nature — team members expressed genuine commitment to producing quality courseware and frustration with PO-related revision cycles. The causes are structural and knowledge-based.

5.1.1 Cause 1: No Formal CFITES PO Writing Training on Onboarding

The FAcT programme onboarding process included a one-day CFITES orientation briefing (n=3 attendees) and access to the CFITES volumes on a shared drive, but no structured instruction in PO writing methodology. Team members reported learning PO writing through trial and error, informal peer review, and audit feedback — all after-the-fact correction rather than front-loaded skill development.

> *Interview, Courseware Developer (7 years experience): "I didn't know there was a specific formula until a DND reviewer sent back a correction. Nobody told me what the three parts were supposed to be. I'd been writing POs for six months."*

5.1.2 Cause 2: Confusion Between CFITES Cognitive Levels and Bloom's Taxonomy

Several team members had prior adult education or instructional design training that introduced Bloom's Taxonomy. The CFITES cognitive framework — Recall, Apply (Understand), and Problem-Solve — uses different level labels and a condensed three-level structure. Three of seven team members reported applying Bloom's verb lists to CFITES POs, producing level mismatches that caused audit flags.

> *Interview, Instructional Designer (B.Ed., 3 years experience): "I was using 'Bloom's verbs' — I had a chart on my wall. Nobody had shown me the CFITES version. I thought they were the same thing."*

5.1.3 Cause 3: SME-Driven PO Drafting Without ID Oversight

In several observed instances, QFI SMEs drafted PO language directly and handed it to IDs for formatting into the Training Plan template, bypassing structured ID review. QFIs are aviation experts with deep task knowledge but typically no formal CFITES PO writing training. Their natural language for task description tends toward outcome-oriented statements ('students need to know how to read a METAR') rather than CFITES-structured performance statements.

> *Observation note: In a courseware development meeting, an SME submitted six PO drafts. Four used 'understand' or 'know' as the primary verb. The ID accepted them without revision and proceeded to lesson planning.*

5.1.4 Cause 4: Absence of a Diagnostic Reference Tool at Point of Work

No job aid, checklist, or quick reference guide exists within the FAcT programme to support PO writing or review. Team members rely on CFITES Volume 4 (207 pages) as their sole reference — a document that is comprehensive but not calibrated for quick consultation during active courseware development. The absence of a

concise, actionable point-of-work reference is a contributing factor to both the incidence of non-compliant POs and the length of revision cycles.

6. RECOMMENDED INTERVENTION

6.1 Intervention Rationale

Given the causal analysis, the gap requires an intervention that delivers structured, applied instruction in CFITES PO writing — specifically the three-component framework — and provides a point-of-work reference tool for ongoing use. The gap is not attitudinal, motivational, or resource-constrained; it is a knowledge and skill gap that responds directly to well-designed training.

A self-directed eLearning module is the recommended delivery modality for the following reasons:

Consideration	Rationale for eLearning
Geographic distribution	FAcT ID team is split between Portage la Prairie MB and Ottawa ON; synchronous delivery requires significant logistics and cost for a team of 7.
Scope of content	The target skill — CFITES PO writing at the three-component level — is well-bounded and suited to a 15–20 minute intervention. A full-day workshop would be disproportionate.
Repeatability	New courseware developers will be onboarded over the life of the FAcT contract. A self-directed module can be assigned to each new hire without instructor scheduling.
DND compliance evidence	A completion record from an eLearning LMS or portfolio link satisfies the CAR requirement for documented training delivery.
Portfolio signal	If developed in Articulate Rise 360 and published via shareable link, the module also functions as a demonstrable artefact of instructional design practice for programme documentation and stakeholder communication.

6.2 Intervention Specification

Parameter	Specification
Module title	Writing Performance Objectives the CFITES Way
Terminal learning objective	Given a job task description drawn from a military aircrew training context, the learner will write a complete CFITES-aligned Performance Objective — including a correctly specified Action, Conditions, and Standard — with no errors in any of the three components.
Enabling objectives	EO 1: Explain why Performance Objectives are the load-bearing element of the CFITES framework (Recall) EO 2: Identify the three required components of a CFITES PO and the criteria each must meet (Recall) EO 3: Select appropriate observable, measurable verbs mapped to CFITES cognitive levels (Apply) EO 4: Diagnose non-compliant POs by identifying the component that is absent or deficient (Apply) EO 5: Construct a three-component PO from a job task description (Apply)

Parameter	Specification
Target population	SkyAlyne FAcT courseware developers and instructional designers; secondary: QFI SMEs who draft or review POs
Delivery modality	Self-directed eLearning — Articulate Rise 360, published to shareable web link
Estimated learner time	15–20 minutes
Key interactions	Labeled graphic — ACTION | CONDITIONS | STANDARD diagram with descriptor reveals Accordion — three-component drill-down with verb taxonomy tables Sorting activity — diagnose 6 sample POs as well-formed / fixable / broken Branching scenario — construct a PO from a QFI job task description in three scaffolded steps
Job aid	One-page CFITES PO Writing Checklist (downloadable PDF) provided within the module
Evaluation	Module completion record (Rise 360); diagnostic sorting activity score; scenario completion rate. Recommend 90-day post-training artefact review against the same 23-PO sample to confirm transfer.
Development estimate	15–25 hours (including SME review of content, graphic asset creation, Rise 360 build, and review iteration)

6.3 Content Scope Rationale

The recommended module addresses PO writing at the three-component level only. It does not attempt to cover the full CFITES six-phase model, Task Analysis methodology, or Training Plan construction. This scope decision is deliberate:

• The causal analysis identified PO component knowledge as the proximate skill gap. Broader CFITES literacy training, while valuable, is a separate need that should be addressed through a subsequent intervention if indicated.

• A tightly scoped 15–20 minute module maximizes completion rates among a production-focused workforce and minimizes the scheduling impact on active FAcT development timelines.

• The DND Corrective Action Request is specifically tied to PO writing compliance. A scoped intervention directly addresses the documented compliance obligation.

7. IMPLEMENTATION CONSIDERATIONS

7.1 Suggested Development Timeline

Phase	Activity	Owner	Duration
Design	Review TNA; confirm terminal and enabling objectives with Training Manager and DND Standards Officer; finalize module scope	ID Lead	1 week
Content development	Draft script; SME review of CFITES content accuracy (QFI); revise	ID + QFI SME	1 week
Storyboard	Develop frame-by-frame storyboard with interaction specs; client review	ID Lead	3–4 days
Asset creation	Create Canva graphics (CFITES cycle diagram, three-box diagram, PO checklist PDF); source images	ID Lead	2 days
Rise 360 build	Build module in Articulate Rise 360; internal QA review	ID Lead	3–4 days
Pilot test	Module reviewed by 2 team members for clarity, accuracy, and interaction function; revisions incorporated	ID Lead + reviewers	3 days
Publish and deploy	Publish to Articulate Review 360 or LMS; distribute completion link to all team members; record completions	Training Manager	1 day
Evaluation	90-day post-training PO artefact review (sample of 20 POs) against compliance criteria from this TNA	ID Lead + DND Standards Officer	At 90 days

7.2 Success Metrics

Metric	Target	Measurement Method
Module completion rate	100% of current FAcT ID team within 30 days of deployment	Rise 360 / LMS completion records
Sorting activity accuracy (Section 3)	≥80% of learners correctly classify all 6 POs on first attempt	Rise 360 quiz analytics
Scenario completion (Section 4)	≥90% of learners complete all 3 scenario steps with correct answer selected	Rise 360 scenario analytics

Metric	Target	Measurement Method
Post-training PO compliance (90-day)	≥80% of team-authored POs contain all three compliant components (up from 17% baseline)	Artefact analysis against TNA rubric
DND CAR closure	Formal CAR closure confirmed by DND Directorate of Training and Education	Written confirmation from DND

7.3 Risks and Mitigations

Risk	Likelihood	Impact	Mitigation
SME availability for content review	Medium	Medium — content errors could cause DND rejection	Schedule SME review session as fixed deliverable in project plan; provide marked-up draft, not blank review request
Rise 360 licence availability	Low — SkyAlyne has Articulate 360 licence	High — blocks build phase entirely	Confirm licence access before design phase commences
Scope creep (stakeholders request full CFITES module)	Medium	High — would extend timeline by 4–6 weeks	Anchor scope to CAR compliance requirement; document out-of-scope items as future phase
Transfer failure (learners pass module but do not apply PO writing skills)	Medium	High — gap persists; CAR not meaningfully resolved	Include 90-day post-training artefact review as formal evaluation step; brief line managers on coaching cues
New ID hires not completing module before beginning PO work	Low–Medium	Medium	Embed module in formal onboarding checklist; Training Manager tracks completion

8. DOCUMENT APPROVAL

This Training Needs Assessment is submitted for review and approval by the following stakeholders. Approval authorizes commencement of the Design phase for the recommended eLearning intervention.

Stakeholder / Role	Name	Signature	Date
Instructional Designer (Author)	Jordan P.R. Smith, PhD		
Training Manager, FAcT	[Name]		
Head, Training, SkyAlyne	[Name]		
DND Standards Officer	[Name / Rank]		

APPENDIX A — Reference Documents

- A-P9-050-000/PT-006 — CFITES Volume 6: Conduct of Instructional Programmes (Director of Army Training, 2017)
- A-P9-050-000/PT-004 — CFITES Volume 4: Design of Instructional Programmes
- A-P9-050-000/PT-002 — CFITES Volume 2: Analysis of Occupational Requirements
- FAcT Programme Training Plan (PTP) v2.1 — SkyAlyne Canada, 2024 (UNCLASSIFIED)
- DND Q4 2025 FAcT Compliance Audit Report — Directorate of Training and Education (DGTE 3-5)
- DND Corrective Action Request (CAR) #FAcT-2025-14 — CFITES PO Writing Non-Conformance
- Treasury Board Secretariat: Guide to Training Needs Analysis for Federal Programmes (TBS, 2022)

END OF DOCUMENT